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Time Series Prediction with GCTAF and Supervised Contrastive Learning

Can missing-aware temporal representation learning along with supervised contrastive loss improve solar flare prediction under class imbalance?

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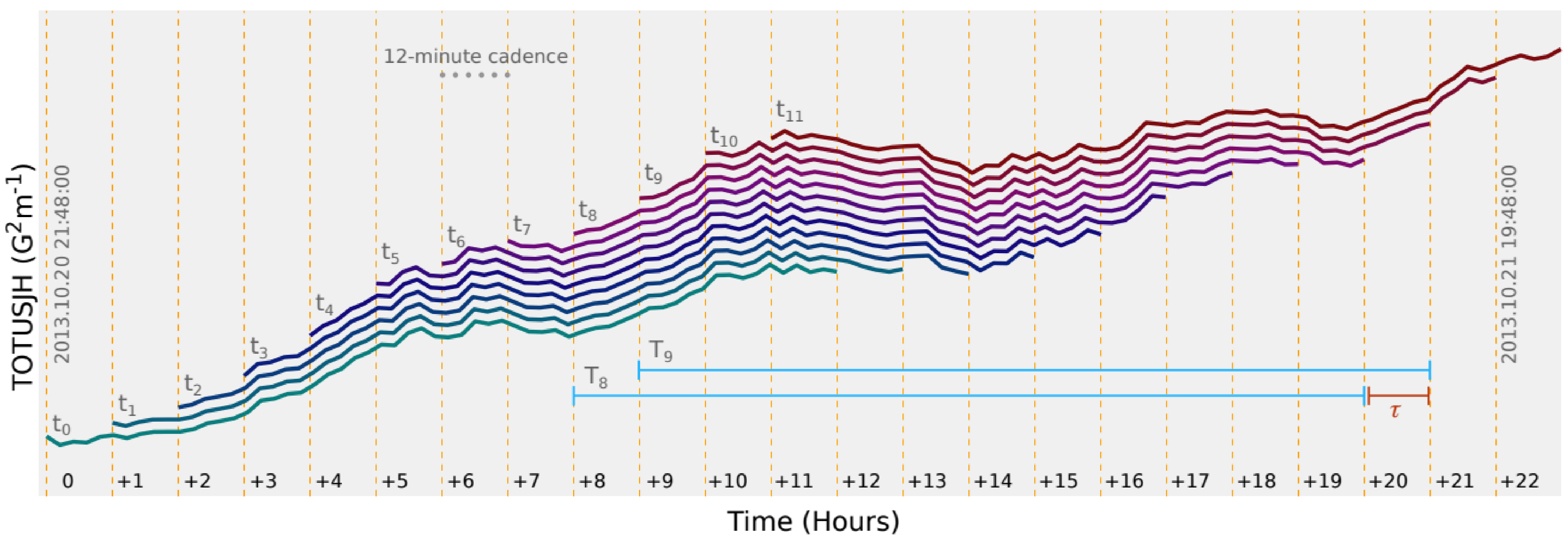
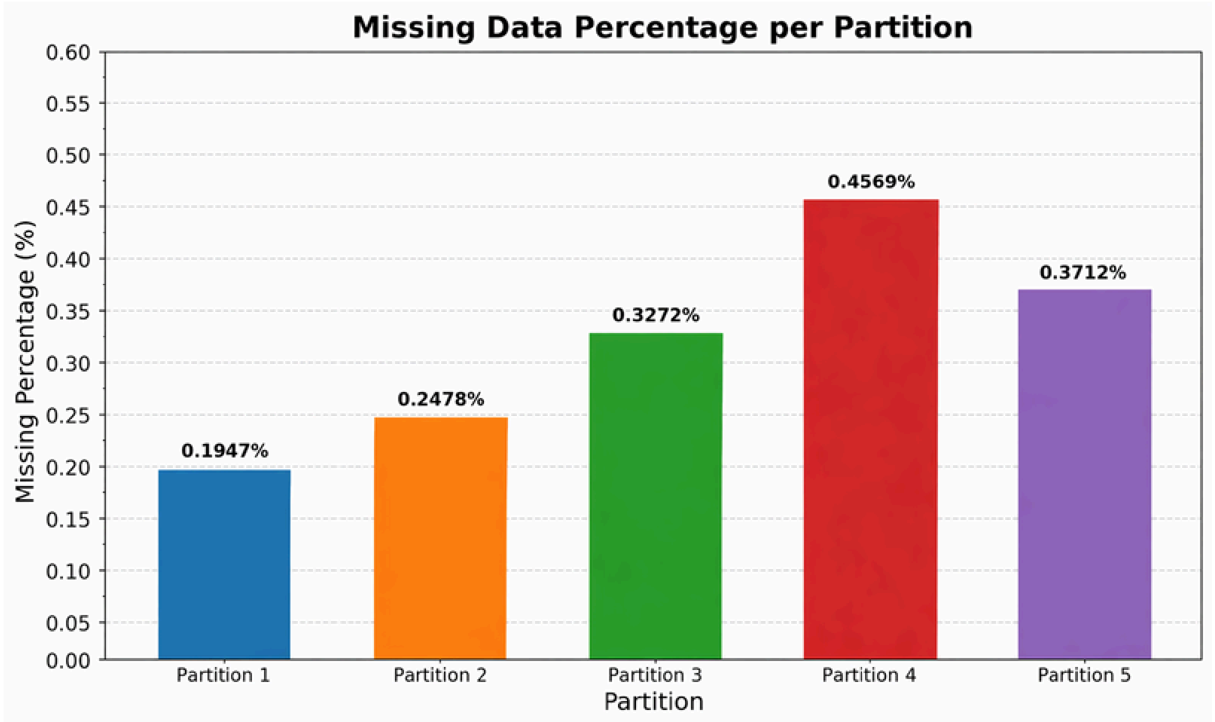
About SWAN-SF Dataset

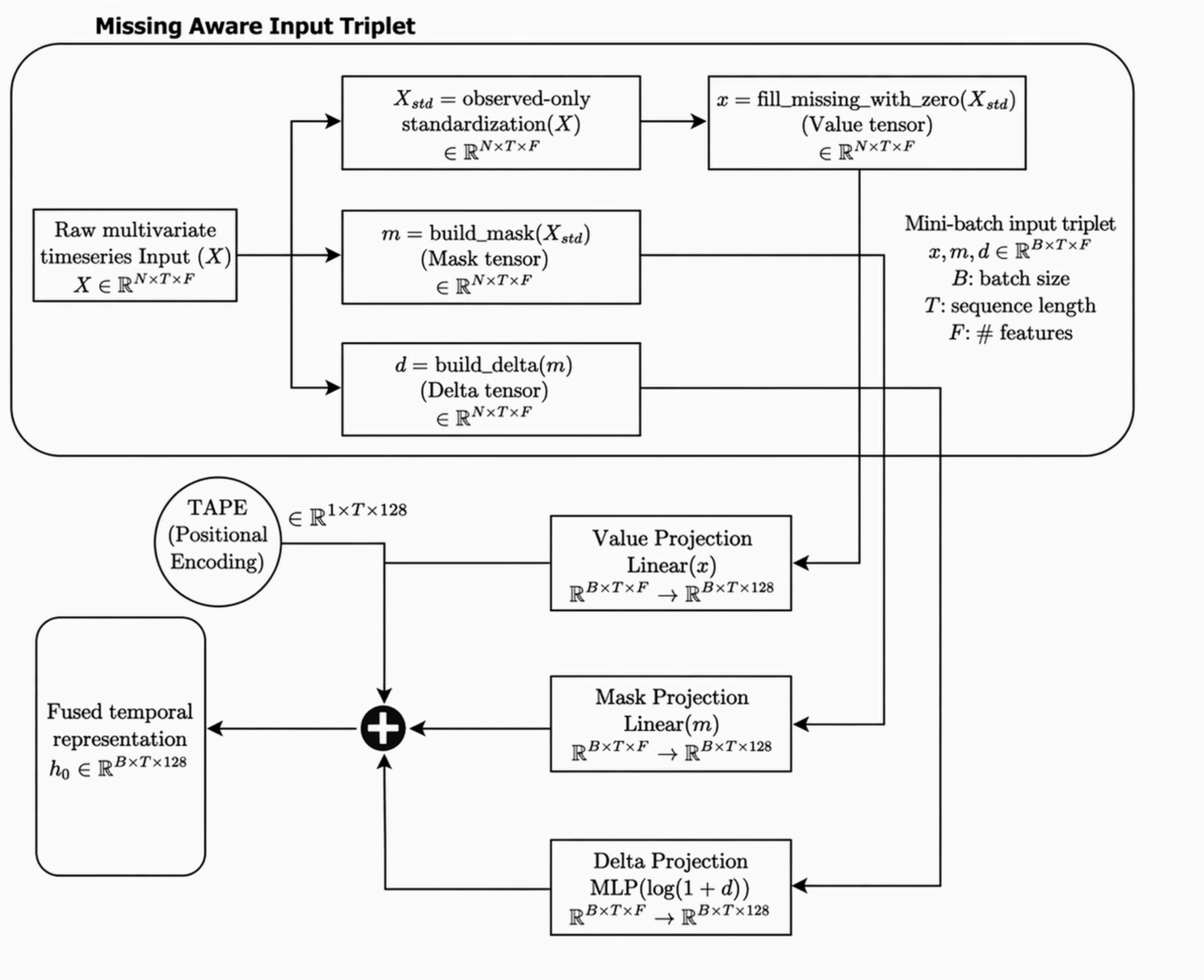
- Swan-SF dataset
  - Class Imbalance
  - Missing Values
  - Temporal Coherence

How to Train Your Flare Forecasting Models





Data Preprocessing: Missing-Aware Triplet Construction for temporal representation



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Architecture : SWAN Encoder with Supervised Contrastive Learning and Linear Classification

Algorithm 1 High-Level Pipeline of the SWAN-Style Flare Prediction Model

Require: Multivariate time series $X \in \mathbb{R}^{T \times F}$
Ensure: Binary flare prediction $\hat{y}$

Construct mask M indicating observed and missing entries
Compute delta matrix D representing time since last valid observation
Standardize input features and replace missing entries with zeros
Project inputs into latent space:

$$X_v = W_x X, \qquad X_m = W_m M, \qquad X_d = \mathrm{MLP}(\log(1+D))$$

Add learned temporal embedding:

$$H^{(0)} = X_v + X_m + X_d + E_{time}$$

for encoder block $\ell = 1$ to L **do**
  Let the input sequence representation be $H^{(\ell-1)} \in \mathbb{R}^{T \times d}$
  Initialize learnable global tokens $G_0^{(\ell)} \in \mathbb{R}^{K \times d}$
  Compute global-token cross-attention over the sequence:

$$\tilde{G}^{(\ell)} = \mathrm{MHA}\left(Q = G_0^{(\ell)},\ K = H^{(\ell-1)},\ V = H^{(\ell-1)}\right)$$

  Aggregate the updated global tokens into a summary vector:

$$g^{(\ell)} = \frac{1}{K} \sum_{k=1}^{K} \tilde{G}_k^{(\ell)}$$

  Inject the global summary into each temporal token:

$$\hat{H}^{(\ell)} = H^{(\ell-1)} + \mathbf{1}_T \left(g^{(\ell)}\right)^\top$$

  where $\mathbf{1}_T \in \mathbb{R}^T$ replicates $g^{(\ell)}$ across all T time steps
  Apply layer normalization:

$$\overline{H}^{(\ell)} = \mathrm{LN}\left(\hat{H}^{(\ell)}\right)$$

  Compute relative-position-aware self-attention:

$$A_{ij}^{(\ell)} = \frac{\left(Q_i^{(\ell)}\right)\left(K_j^{(\ell)}\right)^\top}{\sqrt{d_h}} + b_{ij}^{(\ell)}, \qquad b_{ij}^{(\ell)} = \mathrm{MLP}\left(\mathrm{sign}(i-j)\log(1+|i-j|)\right)$$

$$H^{(\ell)} = \mathrm{MHA}\left(\overline{H}^{(\ell)}\right)$$

end for
Apply layer normalization to final sequence representation
Compute attention pooling over time to obtain a fixed-length embedding h

Training Phase 1: Use projection head and supervised contrastive loss to learn discriminative embedding space
Training Phase 2: Freeze encoder and train linear classifier on h
Inference: Use trained classifier to predict flare/non-flare label

$$X_v = W_x X$$

$$X_m = W_m M$$

$$X_d = \mathrm{MLP}(\log(1+D))$$

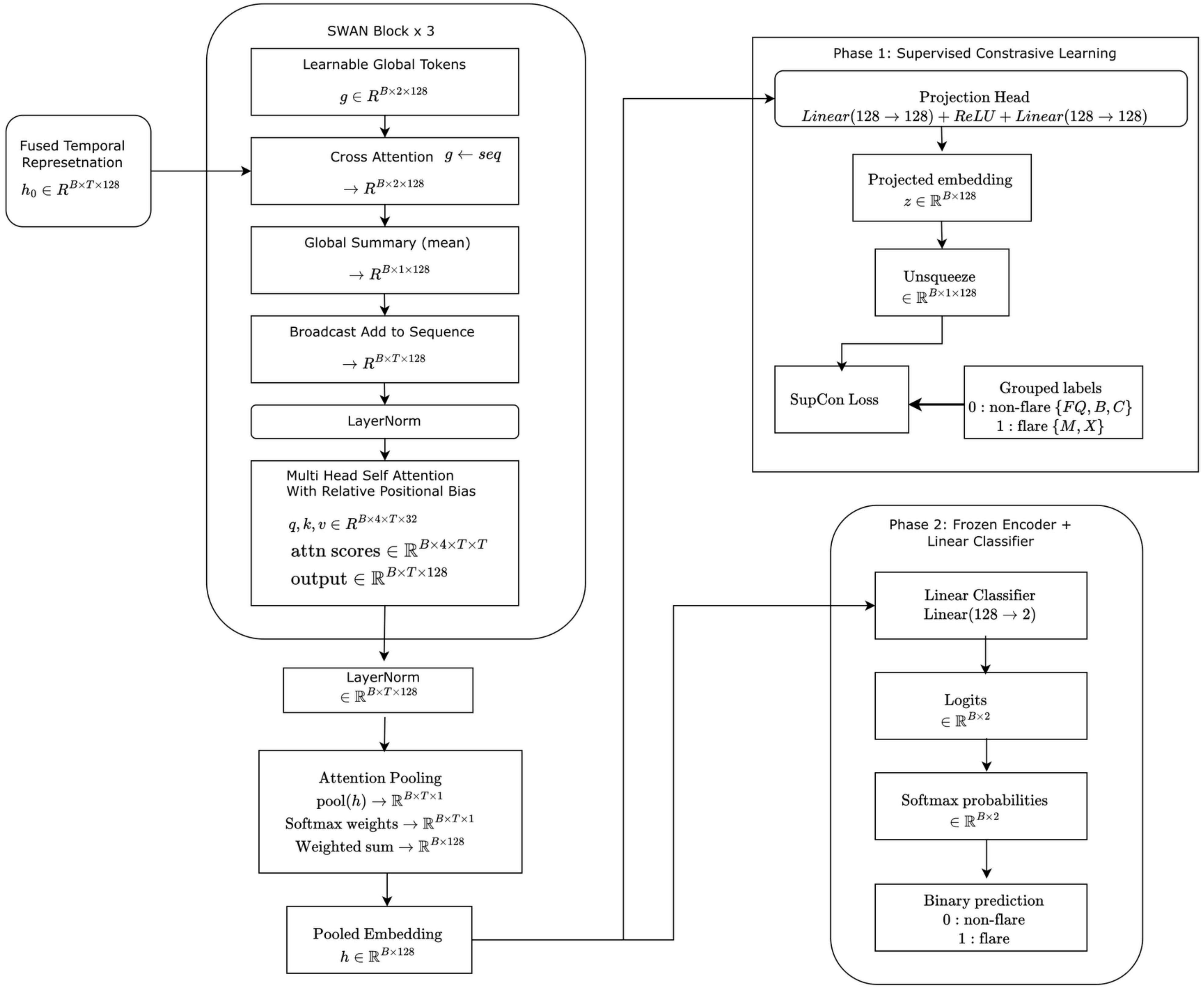
$$H^{(0)} = X_v + X_m + X_d + E_{\text{time}}$$

Formula	Annotation
$H^{(0)} = X_v + X_m + X_d + E_{\text{time}}$	Build initial temporal representation from values, missingness, time gaps, and temporal encoding.
$\tilde{G}^{(\ell)} = \mathrm{MHA}(Q = G_0^{(\ell)}, K = H^{(\ell-1)}, V = H^{(\ell-1)})$	Global tokens attend to the full time series.
$\hat{H}^{(\ell)} = H^{(\ell-1)} + \mathbf{1}_T (g^{(\ell)})^\top$	Inject global context back into every time step.
$A_{ij}^{(\ell)} = \frac{Q_i^{(\ell)}(K_j^{(\ell)})^\top}{\sqrt{d_h}} + b_{ij}^{(\ell)}$	Temporal self-attention with relative-position bias.

$$\mathcal{L}_{\text{supcon}} = \sum_{i=1}^{N} \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(z_i^\top z_p/\tau)}{\sum_{a \neq i} \exp(z_i^\top z_a/\tau)}$$

where $P(i) = \{p \neq i : y_p = y_i\}$, z_i is the normalized embedding, and τ is the temperature.

Architecture : SWAN Encoder with Supervised Contrastive Learning and Linear Classification



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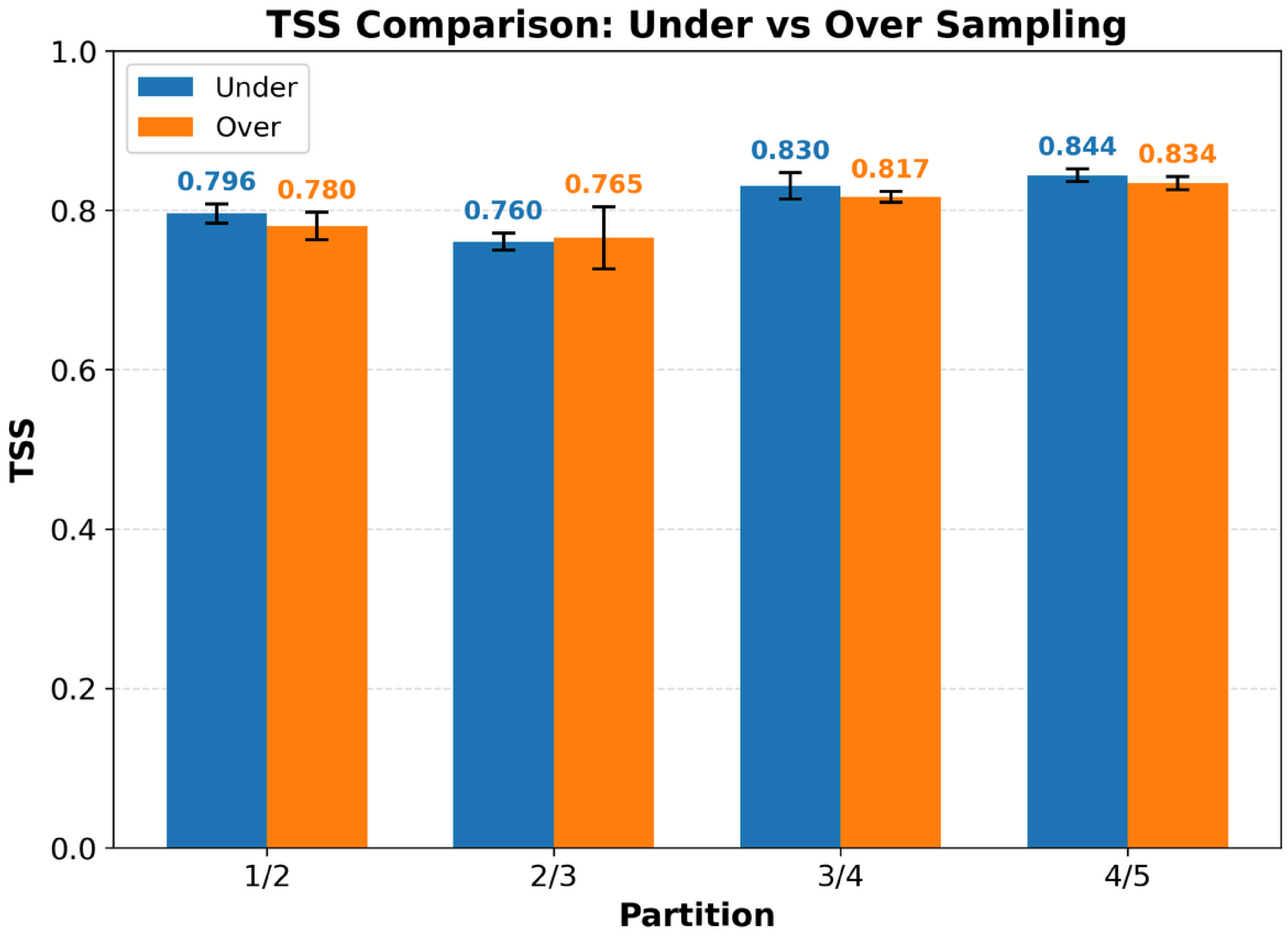
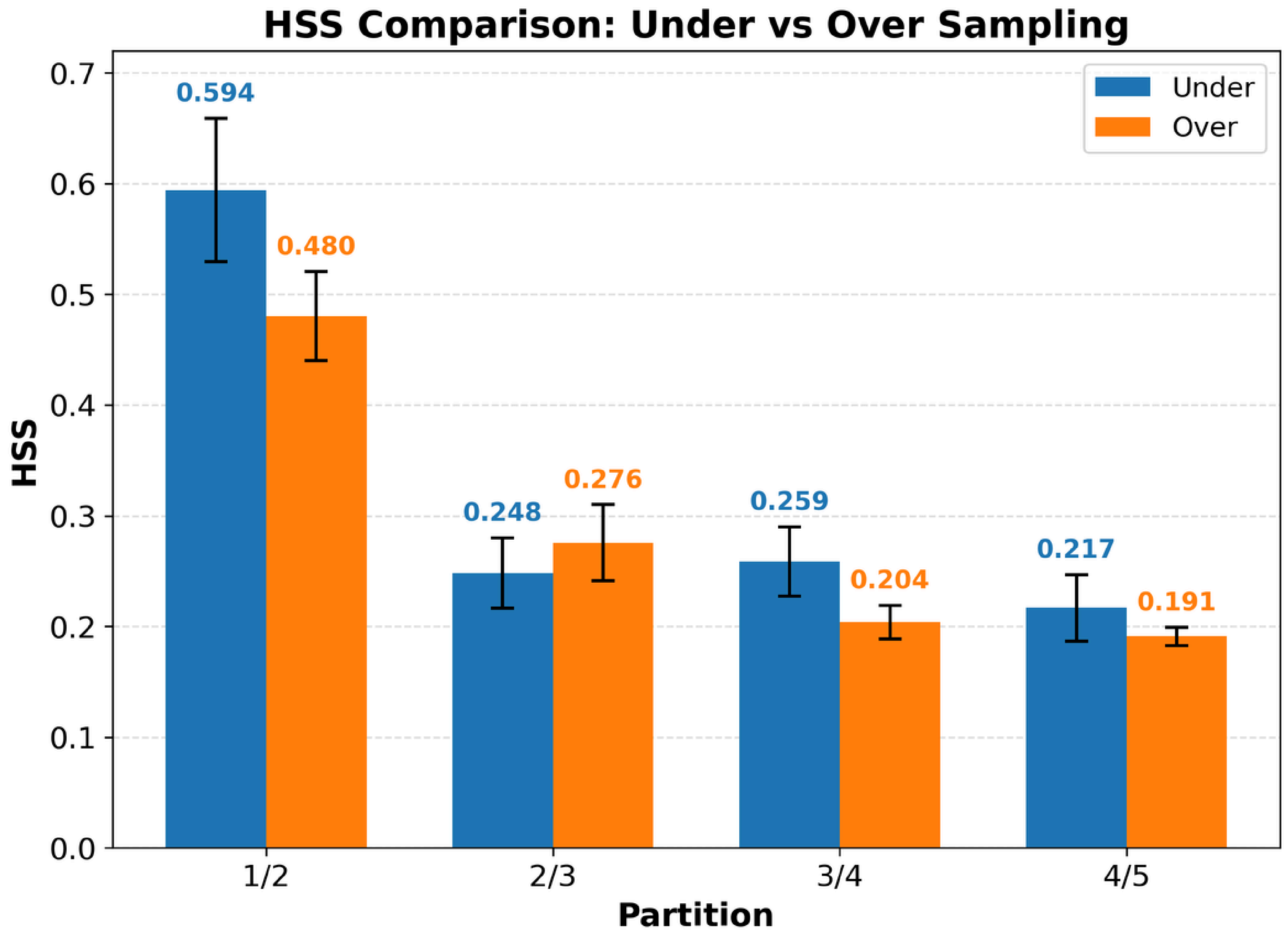
Impact of Sampling Technique on TSS and HSS Across Train/Test Partitions





$$\mathrm{TSS} = \frac{TP \cdot TN - FP \cdot FN}{(TP + FN)(FP + TN)}$$

How well can the model separate positives from negatives?

$$\mathrm{HSS} = \frac{2(TP \cdot TN - FP \cdot FN)}{(TP + FN)(FN + TN) + (TP + FP)(FP + TN)}$$

How much better is this model than random guessing?

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SupCon Loss Transforms Fused Temporal Features into a More Discriminative Latent Space

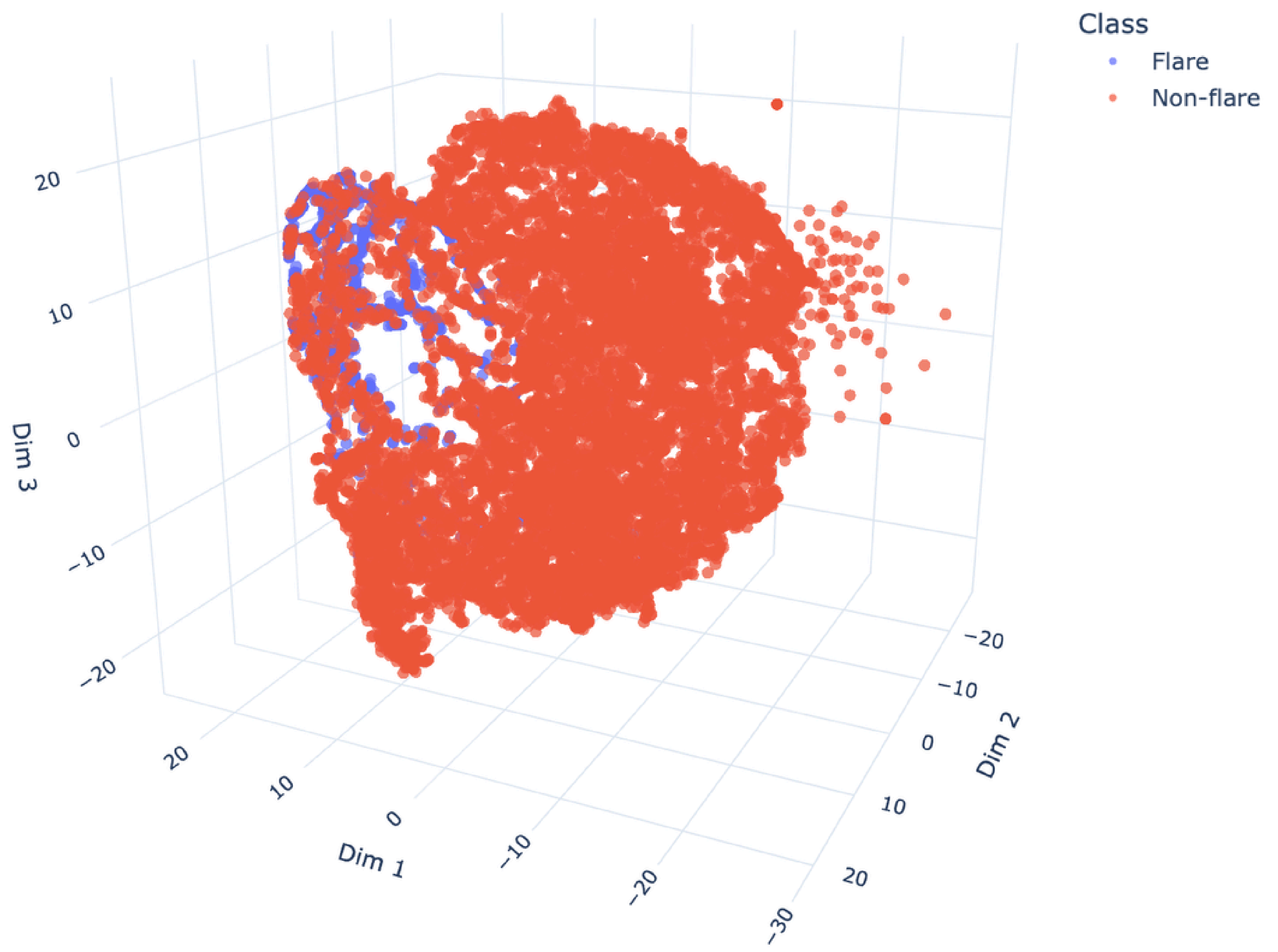

Figure. 3D t-SNE visualization of the fused temporal representations obtained from the Missing-Aware Input Triplet Projection, showing the distribution of samples in the latent space and the degree of separation between classes.

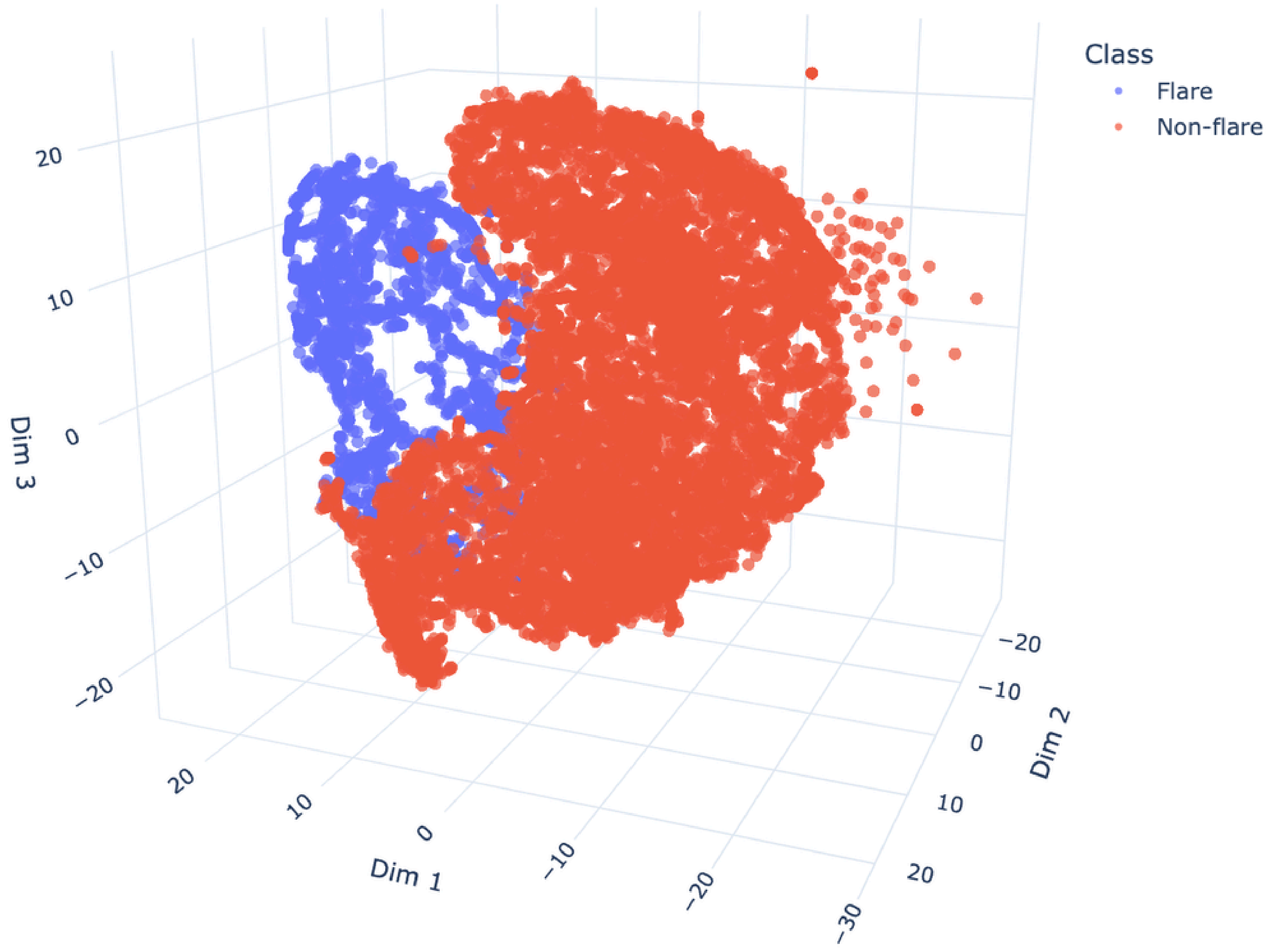

Figure. 3D t-SNE visualization of the projected embeddings z obtained after the projection head in the supervised contrastive learning phase. The plot shows the distribution of samples in the contrastive embedding space and illustrates the degree of class separation between flare and non-flare samples.

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Major Achievement

Comparable Performance

- Undersampling achieves comparable performance with oversampling.
- Eliminates the need for synthetic data generation or duplication.
- Lower computational overhead → faster training and reduced memory usage.
- Practical implication: resource-efficient training without accuracy trade-off.

Strong Predictive Skill (HSS & TSS)

All experiments show:

- Heidke Skill Score (HSS) > 0.19
- True Skill Statistic (TSS) > 0.75

Indicates:

- Reliable classification beyond random chance.
- Strong capability in handling class imbalance.

Leveraging Missing Data

- Missing data is not just noise, it can act as an informative signal.
- Transformer models can:
  - Learn temporal gaps and irregular patterns.
  - Capture implicit structure from missingness.

Ablation

Table 1: Ablation study on the train(P2)/test(P3) undersampled setting.

Variant	**TSS**	**HSS**
Full model	0.7605 ± 0.0106	0.2483 ± 0.0318
w/o GCTAF	0.7276 ± 0.0163	0.2758 ± 0.0134
Relative $\rightarrow$ Positional Encoding	0.7207 ± 0.0271	0.2417 ± 0.0023
w/o TAPE	0.7574 ± 0.0048	0.2345 ± 0.0037
Mask/Delta $\rightarrow$ Median Imputation	0.7503 ± 0.0177	0.2163 ± 0.0048
Group(0,1) $\rightarrow$ Group(FQ,B,C,M,X)	0.7481 ± 0.0140	0.1973 ± 0.0034

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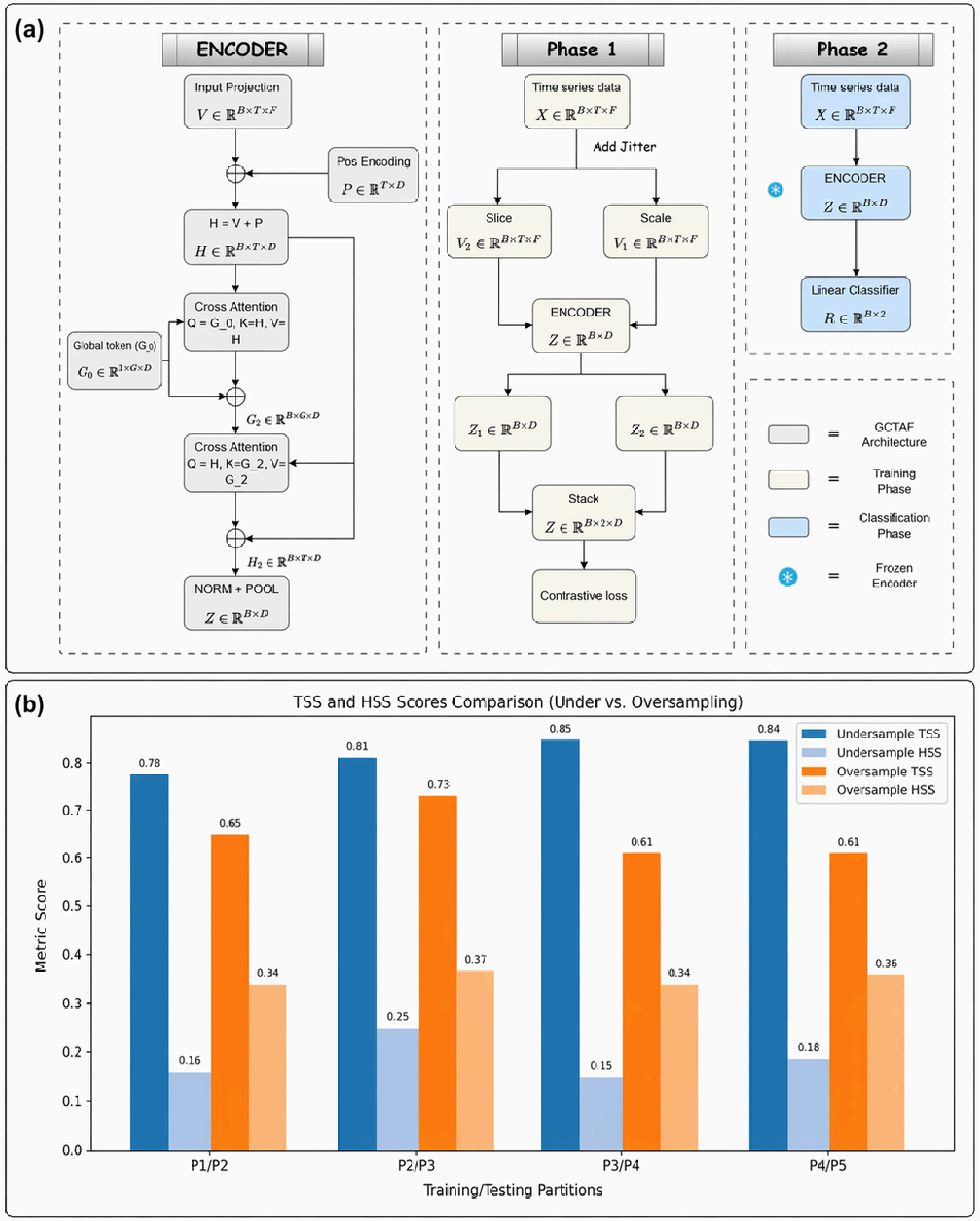
Experiment History

- Two-stage cross-attention did not improve model learning.
- Jitter-based augmentation reduced evaluation performance.
- Missing values were handled using median imputation in previous model.
- AR region information was incorporated but did not made much improvement.



Minor Study - Multivariate Regression Final Model Improves MAE by 10.5% and RMSE by 1.0% over Persistence

Experimental Results

Model	Horizon	Input Length	Test MAE (lower is better)	Test RMSE (lower is better)	Comment
Raw Transformer (Baseline)	12h	60	1.766e21 (vs Pers: -49.7%)	1.384e22 (vs Pers: -117.7%)	Worse than persistence
Delta + Log	24h	60	2.235e21 (vs Pers: +5.6%)	1.996e22 (vs Pers: -23.6%)	MAE better, RMSE worse
Delta + Log + Weighted Loss	24h	60	2.233e21 (vs Pers: +5.7%)	1.586e22 (vs Pers: +1.8%)	Slight improvement
Delta + Log + Longer Context (Final Model)	24h	120	1.056e21 (vs Pers: +10.5%)	6.299e21 (vs Pers: +1.0%)	**Best Model**
Persistence Baseline	24h	—	1.180e21 (Reference)	6.360e21 (Reference)	Reference (naive baseline)

% values show improvement (+) or degradation (-) compared to Persistence Baseline.

Final Model (24h, 120-step) – Feature-wise R^2 on Test Set

Feature	ABSNJZH	SAVNCPP	TOTBSQ	TOTPOT	TOTUSJH	TOTUSJZ	USFLUX	R_VALUE
R^2 Score	0.714	0.686	0.980	0.983	0.961	0.958	0.963	0.660

The final model achieves strong performance on most features; R_VALUE remains the most challenging.

What is Delta Forecasting?

Instead of predicting the absolute future value y_{t+k}, we predict the change (delta) from the last observed value y_t.

$$\Delta y_{t+k} = y_{t+k} - y_t$$

$$\hat{y}_{t+k} = y_t + \Delta \hat{y}_{t+k}$$

y_t: last observed value in input window

y_{t+k}: future value at horizon step k.

Why Use Delta?

- Stabilizes the series and removes strong level/scale effects
- Focuses the model on learning changes (trends, ramps, spikes)
- Usually leads to lower errors and better generalization
- Especially helpful for highly skewed and heavy-tailed variables

Key Takeaway

The 24h transformer with 120-step context + delta forecasting + log transforms delivers the best overall performance, outperforming the persistence baseline.

10.5% lower MAE and 1.0% lower RMSE than persistence.

Conclusion

- Developed a GCTAF-based time-series model for solar flare prediction using the SWAN-SF dataset.
- Applied supervised contrastive learning to improve class separation in the latent embedding space.
- Achieved strong predictive skill across train/test partitions:
  - TSS > 0.75
  - HSS > 0.19
- Found that undersampling performed comparably to oversampling, reducing computational cost and avoiding synthetic data duplication.
- Observed that missing values can carry useful temporal information, rather than acting only as noise
- Overall, the model shows that missing-aware temporal representation + supervised contrastive learning can support reliable solar flare prediction under class imbalance and incomplete data.